

Human Development Index (HDI) Prediction Using Machine Learning

Project Title

Human Development Index (HDI) Prediction Using Machine Learning

Abstract

The Human Development Index (HDI) is a globally recognized metric developed by the United Nations Development Programme (UNDP) to evaluate the overall development of a country. It considers three major dimensions of human development: health, education, and standard of living. This project aims to predict the Human Development Group of a country using Machine Learning techniques based on four significant indicators: Life Expectancy at Birth, Expected Years of Schooling, Mean Years of Schooling, and Gross National Income (GNI) Per Capita.

A Random Forest Classifier is trained using historical HDI data to classify countries into one of four categories: Low, Medium, High, and Very High Human Development. The trained model is integrated with a Flask web application that enables users to input development indicators and instantly obtain the predicted HDI group through a simple web interface.

The developed application achieved an accuracy of **94.87%**, demonstrating the effectiveness of Machine Learning in predicting human development levels. This project highlights the practical use of data science and machine learning in solving real-world socio-economic prediction problems.

Problem Statement

Human development depends on multiple interconnected factors such as healthcare, education, and economic growth. Evaluating a country's Human Development Index manually is complex because it requires analyzing several indicators simultaneously. Traditional approaches are time-consuming and may not provide quick insights for decision-making.

The objective of this project is to develop a Machine Learning-based web application capable of accurately predicting the Human Development Group using important socio-economic indicators. This system helps users quickly estimate the development category of a country and demonstrates how predictive analytics can support educational and research purposes.

Features

- Predicts the Human Development Group using Machine Learning.
- Uses the Random Forest Classification algorithm for accurate predictions.
- Interactive Flask web application with a simple and user-friendly interface.
- Accepts four important HDI indicators as input.
- Displays instant prediction results.
- Responsive web interface designed using HTML and CSS.
- Easy deployment using the Render cloud platform.
- Well-organized GitHub repository with complete documentation.
- Achieved an overall prediction accuracy of **94.87%**.

Tech Stack

Programming Language

- Python 3.11

Machine Learning Libraries

- Pandas
- NumPy
- Scikit-learn
- Joblib

Web Framework

- Flask

Frontend Technologies

- HTML5
- CSS3

Development Tools

- Visual Studio Code
- Git
- GitHub

Deployment Platform

- Render

Installation Steps

Step 1

Clone the repository:

```
git clone https://github.com/Ghani-5/HDI_Prediction_Project.git
```

Step 2

Navigate to the project directory:

```
cd HDI_Prediction_Project
```

Step 3

Install all required dependencies:

```
pip install -r requirements.txt
```

Step 4

Run the Flask application:

```
python app.py
```

Step 5

Open your browser and visit:

```
http://127.0.0.1:5000
```

or access the deployed application:

```
https://hdi-prediction-project.onrender.com/
```

Usage Instructions

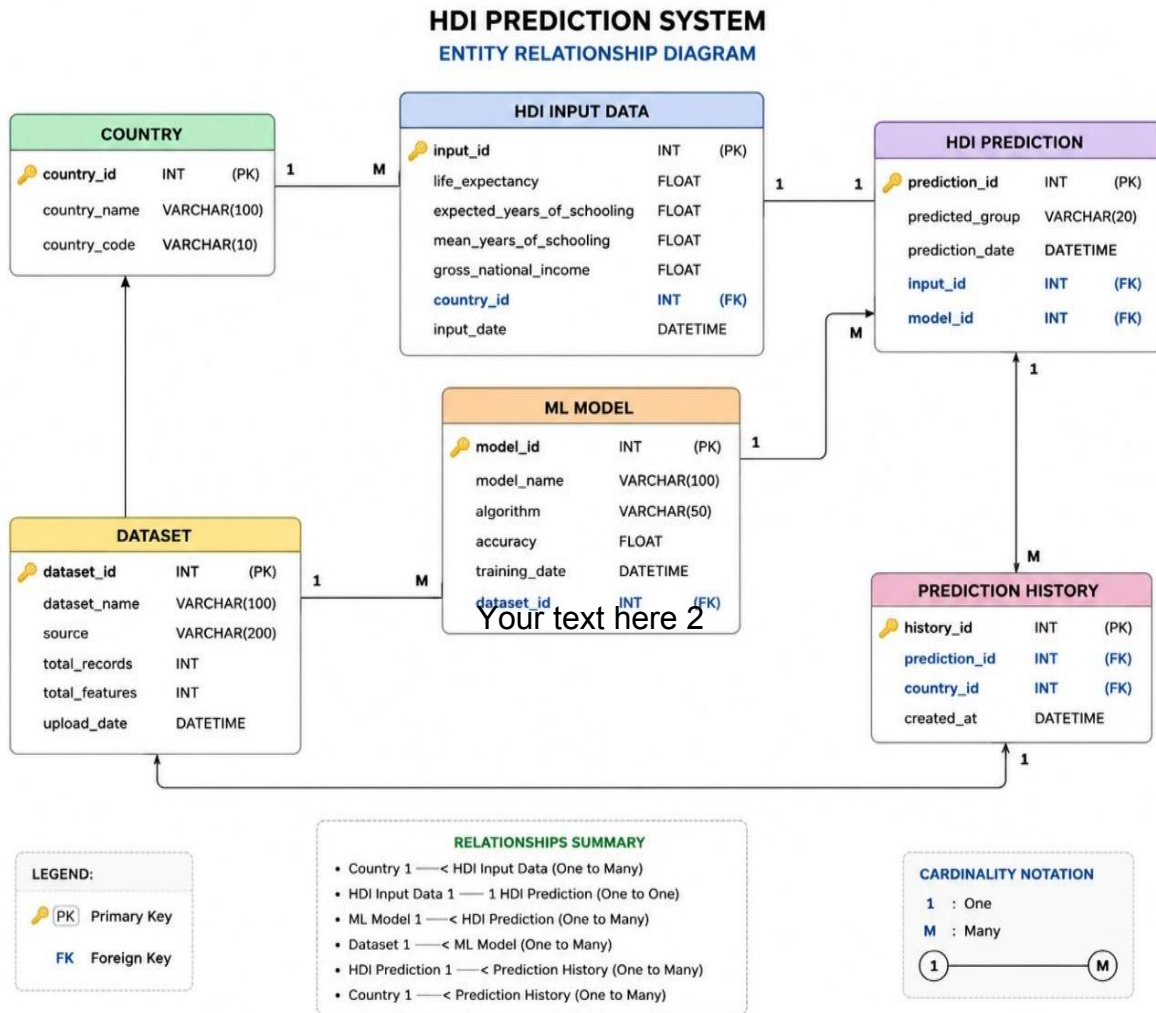
1. Open the HDI Prediction web application.
2. Enter the following input values:
 - Life Expectancy at Birth
 - Expected Years of Schooling
 - Mean Years of Schooling
 - Gross National Income (GNI) Per Capita
3. Click the **Predict** button.
4. The machine learning model processes the input values.
5. The application displays one of the following Human Development Groups:

- Low
- Medium
- High
- Very High

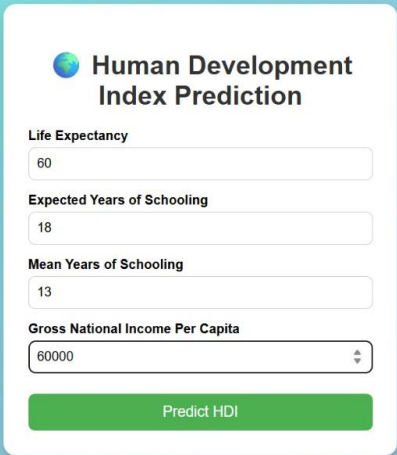
Screenshots

Include the following screenshots:

- Entity Relationship Diagram



- Home Page



The image shows a web form titled "Human Development Index Prediction" with a globe icon. It contains four input fields: "Life Expectancy" with the value 60, "Expected Years of Schooling" with the value 18, "Mean Years of Schooling" with the value 13, and "Gross National Income Per Capita" with the value 60000. A green "Predict HDI" button is at the bottom.

Human Development Index Prediction

Life Expectancy
60

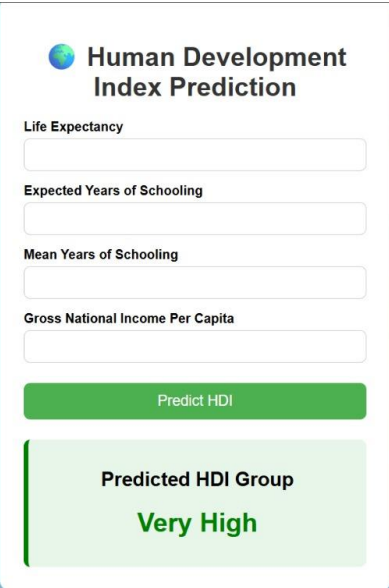
Expected Years of Schooling
18

Mean Years of Schooling
13

Gross National Income Per Capita
60000

Predict HDI

- Prediction Result



This image shows the same form as the previous one, but with the "Predict HDI" button highlighted in green. Below the button, a green box displays the result: "Predicted HDI Group" followed by "Very High" in a larger, bold green font.

Human Development Index Prediction

Life Expectancy

Expected Years of Schooling

Mean Years of Schooling

Gross National Income Per Capita

Predict HDI

Predicted HDI Group
Very High

Future Scope

Although the current application provides accurate HDI group predictions, several enhancements can be implemented in future versions:

- Improve the user interface using Bootstrap or React.
- Support HDI prediction for multiple years.
- Integrate real-time datasets from official sources.
- Add graphical data analysis and visualization dashboards.
- Compare HDI predictions across multiple countries.
- Develop REST APIs for integration with external systems.
- Deploy the application using Docker and cloud services for improved scalability.
- Include advanced machine learning models for comparison and performance optimization.

Team Member Details

S.No	Name	Role
1	SHAIK GHAN SAIDA	Machine Learning Developer & Flask Application Developer (Team Lead)
2	VIDYALATHA GORANTLA	Machine Learning Developer & Flask Application Developer (Team Member)
3	REVANTH GAMINI	Machine Learning Developer & Flask Application Developer (Team Member)
4	MALLAMPATI BHARADWAJ	Machine Learning Developer & Flask Application Developer (Team Member)
5	SHAIK BOPPUDI ABDUL RAHIM	Machine Learning Developer & Flask Application Developer (Team Member)